Global COVID-19 Data Analysis, Visualization, and Forecasting

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Executive Summary

This project analyzes the global spread of COVID-19 using data collected from 187 countries between 22 January 2020 and 27 July 2020. The dataset contains 49,068 records and includes key indicators such as confirmed cases, deaths, recoveries, active cases, and WHO region classifications.The primary objective of the project was to understand the progression of the pandemic, identify the most affected countries and regions, analyze recovery and mortality trends, and forecast future confirmed cases using time-series modeling techniques.The dataset was cleaned and processed using Python and Pandas, followed by Exploratory Data Analysis (EDA) to study confirmed cases, daily new infections, recovery rates, and death rates. Additional analyses were conducted to compare the impact of COVID-19 across countries and WHO regions. Interactive visualizations were created using Plotly to effectively communicate key trends and insights.The analysis revealed a rapid increase in global confirmed cases, particularly after March 2020, indicating widespread transmission across multiple regions. Recovery rates improved steadily over time, while death rates gradually stabilized and declined. Country-level analysis identified the United States, Brazil, and India as the most affected countries, while the Americas emerged as the most impacted WHO region.To predict future trends, a Facebook Prophet forecasting model was developed using historical confirmed case data. The model forecasted continued growth in global infections and projected confirmed cases to exceed approximately 17.5 million by early August 2020.Overall, this project demonstrates how data analytics, visualization, and forecasting techniques can be applied to real-world healthcare data to generate meaningful insights and support data-driven decision-making.

1. Introduction

COVID-19 was one of the most significant global health crises in recent history, affecting millions of people worldwide and creating major challenges for healthcare systems, economies, and societies. As the pandemic spread rapidly across countries, large volumes of data were generated daily, including information on confirmed cases, deaths, recoveries, and active infections.Analyzing this data is essential for understanding how the disease evolved over time, identifying the most affected regions, and evaluating recovery and mortality trends. Data analytics and visualization techniques help transform raw data into meaningful insights, making it easier to study patterns, compare regions, and communicate findings effectively.In this project, global COVID-19 data from 187 countries between 22 January 2020 and 27 July 2020 was analyzed using Python, Pandas, and Plotly. The study focuses on infection trends, recovery rates, death rates, country-level impacts, and WHO regional distribution.In addition to exploratory data analysis, a forecasting model was developed using Facebook Prophet to predict future confirmed cases. The project demonstrates how data analytics, visualization, and time-series forecasting can be applied to real-world healthcare data to generate valuable insights and support data-driven decision-making.

2. Problem Statement and Objectives

2.1 Problem Statement

The COVID-19 pandemic generated large volumes of daily data related to confirmed cases, deaths, recoveries, and active infections across the world. However, raw data alone does not provide a clear understanding of infection trends, regional impacts, or recovery patterns. This project aims to analyze global COVID-19 data, identify key trends, and develop a forecasting model to predict future confirmed cases.

2.2 Project Objectives

The main objectives of this project are:

- Analyze global COVID-19 trends and infection growth.
- Study recovery and mortality patterns.
- Identify the most affected countries and WHO regions.
- Visualize the geographical spread of COVID-19.
- Forecast future confirmed cases using Facebook Prophet.
- Demonstrate the application of data analytics and forecasting techniques on real-world data.

2.3 Scope of the Project

This project analyzes global COVID-19 data from 22 January 2020 to 27 July 2020. The study focuses on confirmed cases, deaths, recoveries, active cases, and WHO region classifications. Forecasting is performed using historical confirmed case data and is intended for analytical and educational purposes.

3. Dataset Description

The dataset used in this project was provided by Intellipaat as part of the COVID-19 Data Analysis and Forecasting Capstone Project. It contains daily COVID-19 statistics from 187 countries, including confirmed cases, deaths, recoveries, active cases, geographical coordinates, and WHO regional classifications.

3.1 Dataset Overview

Attribute	Details
Dataset Source	Intellipaat Capstone Project Dataset
Records	49,068
Countries Covered	187
Time Period	22 January 2020 – 27 July 2020
Data Type	Time-Series Pandemic Data

3.2 Dataset Features

Feature	Description
Country/Region	Country or region name

Lat	Latitude coordinate
Long	Longitude coordinate
Date	Reporting date
Confirmed	Total confirmed cases
Deaths	Total deaths
Recovered	Total recovered cases
Active	Active cases
WHO Region	WHO regional classification

Note: The Province/State column was removed during preprocessing because more than 70% of its values were missing.

3.3 Dataset Characteristics

The dataset is a time-series dataset containing daily observations from multiple countries. It supports trend analysis, country-wise comparison, regional analysis, geographic visualization, and time-series forecasting, making it suitable for studying the global progression of COVID-19.

4. Data Preprocessing

Before performing analysis and forecasting, the dataset was cleaned and prepared to ensure accuracy and consistency. Several preprocessing steps were carried out to make the data suitable for analysis.

4.1 Initial Inspection

The dataset contained 49,068 records and 10 columns. An initial inspection was performed to examine the dataset structure, identify missing values, and verify data types.

4.2 Handling Missing Values: Missing value analysis revealed that the **Province/State** column contained more than 70% missing values. Since the project focused on global-level analysis rather than state-level analysis, this column was removed from the dataset.

4.3 Data Type Conversion:

The **Date** column was originally stored as an object data type and was converted to datetime format. This enabled time-series analysis, date-based aggregation, and compatibility with the Facebook Prophet forecasting model.

4.4 Global Data Aggregation

The original dataset contained separate records for each country on each date. To analyze global trends, the data was aggregated by date, and worldwide totals for confirmed cases, deaths, recoveries, and active cases were calculated.

The resulting dataset contained:

- Date
- Confirmed Cases
- Deaths
- Recovered Cases
- Active Cases

4.5 Feature Engineering

Additional features were created to support analysis:

New Cases

Daily new cases were calculated using the difference between consecutive days of confirmed cases.

New Cases = Today's Confirmed Cases − Previous Day's Confirmed Cases

New Recovered Cases

Daily recovered cases were calculated using the difference between consecutive days of recovered cases.

New Recovered = Today's Recovered Cases − Previous Day's Recovered Cases

Recovery Rate

Recovery Rate (%) = (Recovered / Confirmed) × 100

Death Rate

Death Rate (%) = (Deaths / Confirmed) × 100

These metrics provided deeper insights into infection growth, recovery trends, and mortality patterns.

4.6 Data Preparation for Forecasting

For forecasting, the aggregated dataset was converted into Prophet's required format:

- **ds** → Date
- **y** → Confirmed Cases

The prepared dataset was then used to train the Facebook Prophet model and generate future predictions.

5. Exploratory Data Analysis

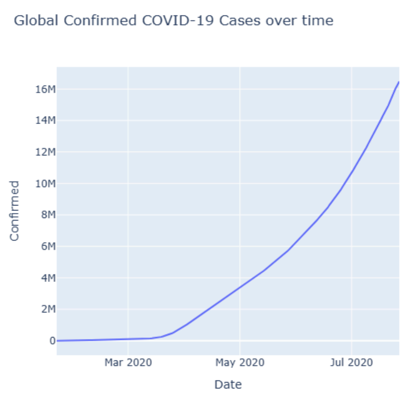
Exploratory Data Analysis (EDA) was performed to understand the progression of COVID-19 and identify key trends in infection, recovery, and mortality patterns. Various visualizations were created using Plotly to analyze the pandemic from different perspectives.

5.1 Global Confirmed Cases Trend

Objective

To analyze the growth of confirmed COVID-19 cases over time.

Figure 1: Global Confirmed Cases Trend



Observations

- Confirmed cases remained low during January and February 2020.
- A rapid increase was observed after March 2020.
- Global confirmed cases exceeded 16 million by July 2020.

Key Insight

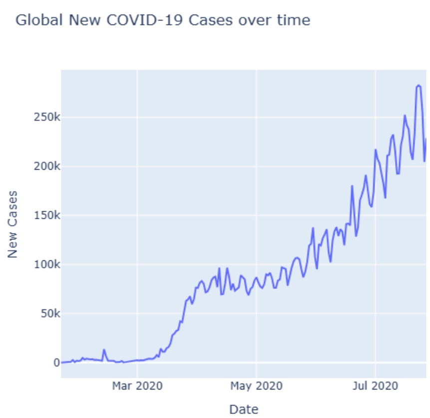
The trend indicates rapid global transmission of COVID-19 and shows that the pandemic was still expanding by the end of the study period.

5.2 Daily New Cases Analysis

Objective

To study the daily growth of COVID-19 infections.

Figure 2: Daily New COVID-19 Cases



Observations

- Daily infections increased significantly from March 2020 onwards.
- Case counts continued to rise throughout the study period.
- Daily new cases exceeded 200,000 by July 2020.

Key Insight

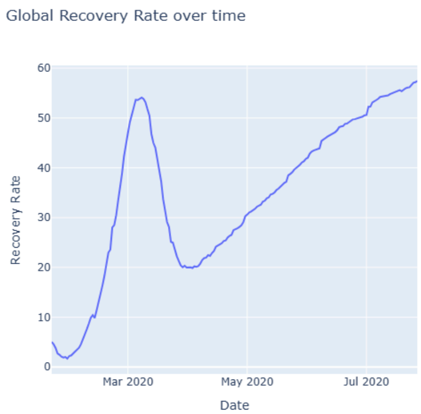
The increasing number of daily infections indicates that COVID-19 transmission remained active and widespread globally.

5.3 Recovery Rate Analysis

Objective

To examine how the recovery rate changed over time.

Figure 3: Recovery Rate Trend



Observations

- Recovery rates increased steadily throughout the study period.
- The global recovery rate exceeded 57% by July 2020.
- Recoveries grew consistently as more patients completed treatment.

Key Insight

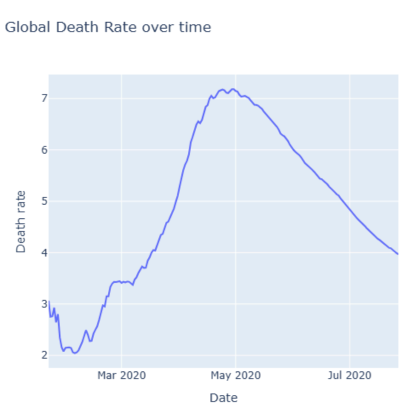
The rising recovery rate suggests improved patient outcomes and healthcare response during the pandemic.

5.4 Death Rate Analysis

Objective

To analyze mortality trends during the pandemic.

Figure 4: Death Rate Trend



Observations

- The death rate increased initially and reached a peak during the middle of the study period.
- A gradual decline was observed afterwards.
- The trend became more stable towards July 2020.

Key Insight

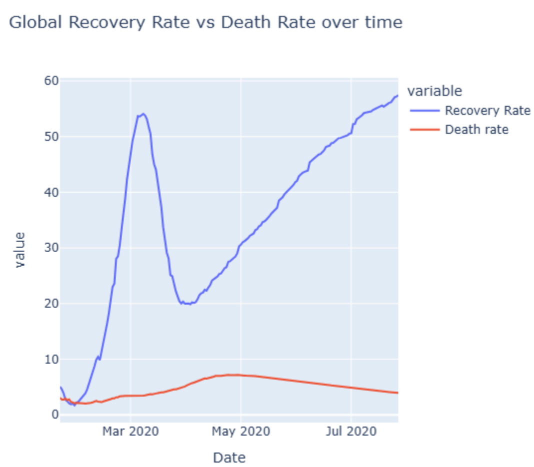
The decline in death rate suggests improvements in medical care, testing, and disease management over time.

5.5 Recovery Rate vs Death Rate Analysis

Objective

To compare recovery and mortality trends throughout the pandemic.

Figure 5: Recovery Rate vs Death Rate



Observations

- Recovery rates increased steadily over time.
- Death rates remained significantly lower than recovery rates.
- The gap between the two metrics widened throughout the study period.

Key Insight : The comparison indicates that recoveries increased at a much faster rate than fatalities, reflecting improved global recovery outcomes.

6. Country-Level Analysis

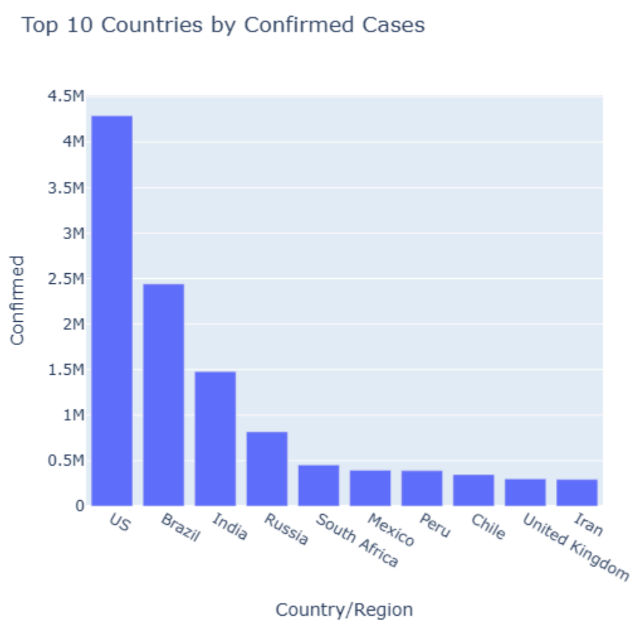
Country-level analysis was performed to identify the countries most affected by COVID-19 based on confirmed cases and deaths.

6.1 Top 10 Countries by Confirmed Cases

Objective

To identify the countries with the highest number of confirmed COVID-19 cases.

Figure 6: Top 10 Countries by Confirmed Cases



Observations

- The United States reported the highest number of confirmed cases.
- Brazil and India were also among the most affected countries.
- A small number of countries accounted for a large share of global infections.

Key Insight

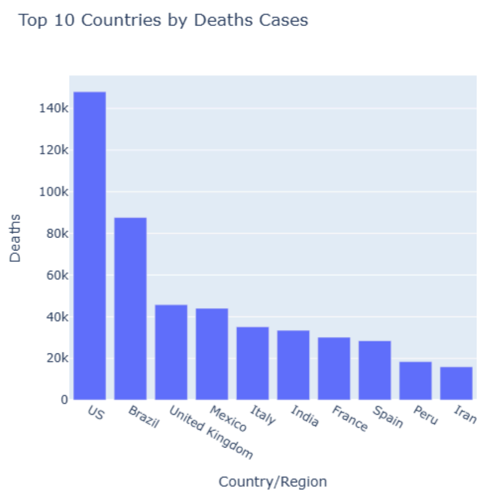
COVID-19 cases were heavily concentrated in a few countries, with the United States, Brazil, and India emerging as the most affected nations during the study period.

6.2 Top 10 Countries by Deaths

Objective

To identify the countries with the highest number of COVID-19-related deaths.

Figure 7: Top 10 Countries by Deaths



Observations

- The United States recorded the highest number of deaths.
- Brazil was the second most affected country in terms of fatalities.
- Countries with high infection counts generally also reported high numbers of deaths.

Key Insight

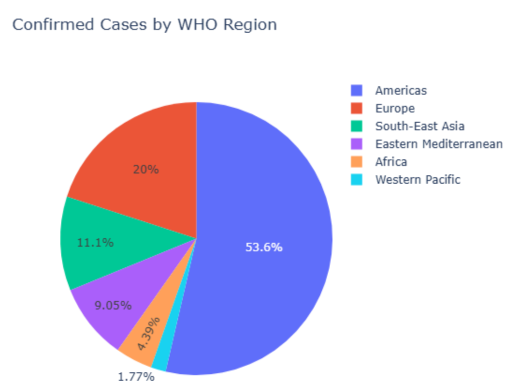
COVID-19 fatalities were concentrated in a limited number of countries, with the United States and Brazil accounting for a significant share of global deaths.

7. WHO Region Analysis

Objective

To analyze the distribution of COVID-19 cases across different WHO regions and identify the most affected regions.

Figure 8: Distribution of Confirmed Cases Across WHO Regions



Observations

- The Americas accounted for the largest share of global confirmed cases.
- Europe and South-East Asia also contributed significantly to total infections.
- Africa and the Western Pacific region reported comparatively fewer cases.
- The distribution of cases was uneven across WHO regions.

Key Insight

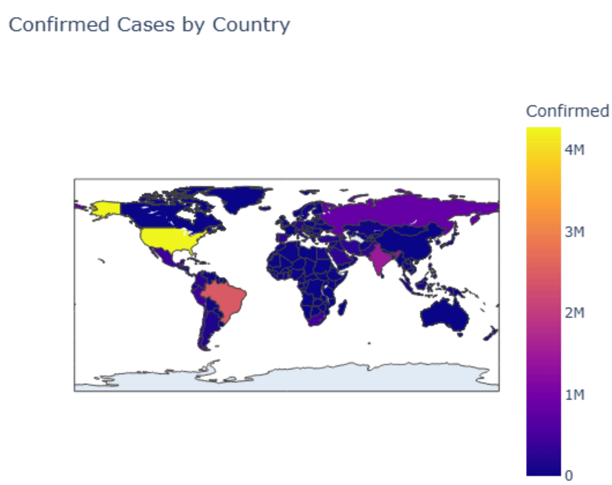
The analysis shows that COVID-19 had a greater impact on certain regions, with the Americas emerging as the most affected WHO region, followed by Europe and South-East Asia.

8. Global Distribution Map

Objective

To visualize the geographical distribution of COVID-19 cases and identify the most affected countries and regions.

Figure 9: Global Distribution of Confirmed COVID-19 Cases



Observations

- COVID-19 cases were reported across nearly all regions of the world.
- The United States, Brazil, and India showed the highest infection levels.
- Several European countries also recorded substantial case counts.
- The intensity of infections varied significantly across countries and regions.

Key Insight

The global distribution map highlights the worldwide spread of COVID-19 while showing that certain countries experienced a much greater infection burden than others.

9. Forecasting Using Facebook Prophet

Objective

To predict future COVID-19 confirmed cases using historical data and demonstrate the application of time-series forecasting techniques.

Methodology

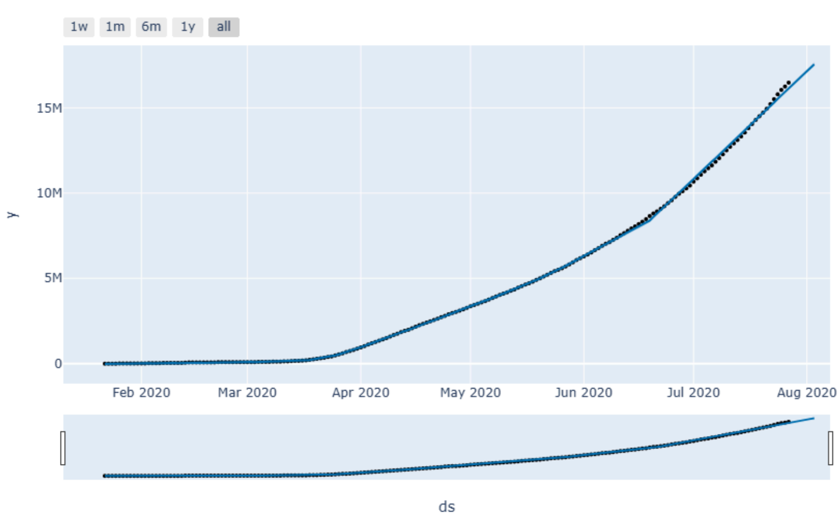
Facebook Prophet was used to build the forecasting model due to its simplicity and effectiveness in handling time-series data. The aggregated global confirmed cases dataset was converted into Prophet's required format:

- **ds** → Date
- **y** → Confirmed Cases

The model was trained on historical data and used to forecast confirmed cases for the next 7 days.

Forecast Visualization

Figure 10: COVID-19 Confirmed Cases Forecast Using Facebook Prophet



Observations

- The forecast shows a continued increase in confirmed COVID-19 cases.

- No significant decline or stabilization is observed during the forecast period.
- The confidence interval widens slightly for future dates, indicating increasing uncertainty.
- The model predicted global confirmed cases to exceed approximately 17.5 million by early August 2020.

Key Insight

The Facebook Prophet model successfully captured the overall growth trend of COVID-19 and predicted continued growth in confirmed cases. The results suggest that the pandemic remained in an active expansion phase at the end of the study period.

10. Key Findings

The key findings of the analysis are summarized below:

- Global confirmed COVID-19 cases increased rapidly, showing an exponential growth trend throughout the study period.
- Daily new infections continued to rise, indicating ongoing global transmission.
- The recovery rate improved significantly, increasing from approximately 5% to over 57% by July 2020.
- The death rate initially increased but gradually stabilized and declined over time.
- Recovery rates grew faster than death rates, indicating improved patient outcomes.
- The United States, Brazil, and India were the most affected countries in terms of confirmed cases.
- The United States and Brazil recorded the highest numbers of COVID-19-related deaths.
- The Americas emerged as the most affected WHO region.
- The global distribution map highlighted the worldwide spread of COVID-19, with varying levels of severity across countries.
- The Facebook Prophet model predicted continued growth in confirmed cases, with global infections expected to exceed 17.5 million by early August 2020.

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11. Conclusion

This project analyzed global COVID-19 data from 187 countries between January and July 2020 using Python, Pandas, Plotly, and Facebook Prophet. The study explored infection trends, recovery patterns, mortality rates, regional impacts, and country-level distributions.

The analysis revealed rapid growth in confirmed cases, improving recovery rates, and a gradual decline in death rates. In addition, a forecasting model was developed to predict future confirmed cases, indicating continued growth in infections during the forecast period.

Overall, the project demonstrates the application of data analytics, visualization, and forecasting techniques to generate meaningful insights from real-world healthcare data.

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12. Future Scope

This project can be extended in several ways:

- Develop forecasting models for deaths, recoveries, and active cases.
- Perform country-specific analysis and forecasting.
- Build an interactive dashboard using Streamlit or Power BI.
- Evaluate model performance using metrics such as MAE, RMSE, and MAPE.
- Compare Prophet with ARIMA, SARIMA, and LSTM models.
- Incorporate additional factors such as population, vaccination rates, and healthcare infrastructure.
- Extend the analysis using more recent COVID-19 datasets.

These improvements can provide deeper insights and enhance forecasting accuracy.